

Tutorial - Week 2

1. Determine which equations define functions of y in terms of x .

- (a) $x + y = 1$.
- (b) $x^2 + y^2 = 1$.
- (c) $x^3 + y^3 = 1$.
- (d) $x^2 + y = 1$.
- (e) $x + y^2 = 1$.
- (f) $x^2 + y^3 = 1$.
- (g) $xy = 1$.
- (h) $xy = -2$.
- (i) $xy = 0$.

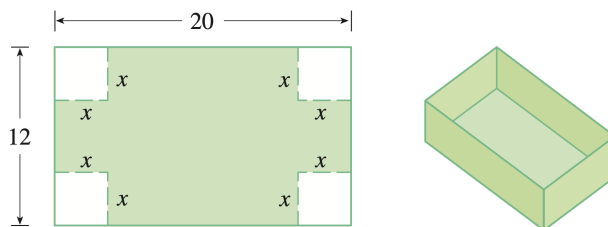
2. Evaluate the difference quotient for the given function. Simplify your answer.

- (a) $f(x) = 4 + 3x - x^3$, $\frac{f(3+h)-f(3)}{h}$.
- (b) $f(x) = 2x^3$, $\frac{f(x_0+h)-f(x_0)}{h}$.
- (c) $f(x) = \frac{1}{x}$, $\frac{f(x)-f(x_0)}{x-x_0}$.

What is this quotient representing?

3. Let $f(x) = x^2 + 1$. Find the line that intersects the graph of f at $x = 1$ and $x = 1 + h$.

4. A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 12 in. by 20 in. by cutting out equal squares of side x at each corner and then folding up the sides as in the figure. Express the volume V of the box as a function of x .



5. Find the domain of the following functions

- (a) $f(x) = \frac{x-1}{x^2+x-2}$.
- (b) $g(x) = \frac{x-3}{x^2-25}$.
- (c) $h(x) = \frac{1-x}{1+\frac{1}{1-x}}$.
- (d) $j(x) = \sqrt{\sin x}$.
- (e) $k(x) = \sqrt{-t-1} + \sqrt{t+2}$.
- (f) $l(x) = \frac{1}{\sqrt{e^x-2}}$.
- (g) $u(x) = \frac{x-1}{\sqrt{x-1}}$.
- (h) $v(x) = \sqrt{\frac{1+\sin x}{1+\cos x}}$.

6. Find the range and period of each of the following functions

(a) $f_1(x) = \sin(2x)$.

(b) $f_2(x) = 2\sin(x)$.

(c) $f_3(x) = \cos(\frac{x}{3})$.

(d) $f_4(x) = 1 - \sin^2(\frac{3x}{4})$.

7. Consider $f(x) = x^2$ with x in $[0, \infty)$. Show that the function f is increasing, that is, if x and y are non-negative real numbers and $x < y$, then $x^2 < y^2$ (Hint: find an expression to multiply in the first inequality).

8. Determine if the following functions are increasing, decreasing, or neither.

(a) $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ such that $f(x)$ is the number of positive prime divisors of x .

(b) $f: [0, 6] \rightarrow [0, +\infty)$ such that $f(x)$ is the time in minutes needed by a bike rider to travel x kilometers along Daxue Road, from the main gate of the North Campus towards the center of Shantou.

9. Let $f(x)$ be the piecewise-defined function as follows:

$$f(x) = \begin{cases} x^2 + 2x & \text{if } x \geq 0 \\ -x^2 + 2x & \text{if } x < 0 \end{cases}$$

(a) Find its range

(b) Determine whether f is increasing, decreasing, or neither.

(c) Determine whether f is even, odd, or neither.

10. Consider the function $f(x) = |x^2 - 16|$.

(a) Find an explicit formula for $f(x)$.

(b) Determine the intervals where the function is increasing or decreasing.

(c) Sketch the function.

11. Rewrite the functions from Exercise 5 in the form of sums, differences, products, quotients or powers of other functions.

12. Show how to obtain the graph of $g(x) = \cos^2 x$ from the graph of $f(x) = \cos(x)$ by shrinking, stretching and/or shifting.

13. Find the functions $f \circ g$ and $g \circ f$, their domains and ranges.

(a) $f(x) = \sin x$, $g(x) = |x|$.

(b) $f(x) = x^2 + x^4$, $g(x) = |x|$.

(c) $f(x) = 2 \ln x$, $g(x) = e^x$.

(d) $f(x) = |2x + 1|$, $g(x) = \sqrt{-x + 5}$.

14. (a) Show that the composite of two increasing functions is increasing.

(b) What happens if the functions are both strictly increasing?

(c) What if they are both decreasing?

15. Identify an interval on which the function f is one-to-one.
- (a) $f(x) = \sin x$.
 - (b) $f(x) = \cos x$.
 - (c) $f(x) = \tan x$.
16. Show that the functions are one-to-one and give a formula for their inverses if possible.
- (a) $f(x) = mx + b$.
 - (b) $f(x) = \begin{cases} 2x + 1 & \text{if } x \leq 1, \\ 2 + \sqrt{x} & \text{if } x > 1. \end{cases}$
 - (c) $f(x) = 1 - \frac{4^{-x}}{2}$.
 - (d) $f(x) = 3 \sin^{-1} x + \pi$.
 - (e) $f(x) = \ln(e^x - 2)$.
17. (a) Show that a strictly increasing function is one-to-one and its inverse is also strictly increasing.
(b) What happens with strictly decreasing functions?
18. Write the function $f(x) = \frac{1}{(x-2)^3+5}$ as a composition of functions, and find its inverse using compositions.

